

B.Sc. (Hons.) Semester I Examination 2024–25 (NEP)
SKILL ENHANCEMENT COURSE (SEC)

Paper Code: STASE11 : Data Analysis Using Excel

Time: 2:30 Hours

Max. Marks: 70

Note: Attempt any FIVE questions. Question No. 1 is compulsory. All questions carry equal marks (14 Marks each).

1. Ages of five children are 2, 5, 8, 3, 4 years and are to be entered in Column A of an Excel sheet using rows 2 to 6 with variable name in Row 1. [14]
 - (a) Prepare a layout representation of such an Excel sheet.
 - (b) Write Excel syntax for: (i) Counting number of observations, (ii) Sum of ages, (iii) Minimum age, (iv) Maximum age, and their returned output values.
2. Explain the syntax =QUARTILE(array, quart). Detail the parameters array and quart. Discuss the shortcut commands CTRL+C, CTRL+V, CTRL+SHIFT+J, CTRL+Z, and CTRL+S in MS Excel. [14]
3. Discuss the features and benefits of Pivot Tables in MS Excel. [14]
4. Suppose the first 11 natural numbers are entered in cells A1:A11. Evaluate the outputs of the following Excel formulas: [14]
 - (i) =AVERAGE(A1:A11)
 - (ii) =VAR.P(A1:A11)
 - (iii) =MEDIAN(A1:A11)
 - (iv) =(MAX(A1:A11) - MIN(A1:A11)) / (MAX(A1:A11) + MIN(A1:A11))
5. Explain the four levels of measurement scales (Nominal, Ordinal, Interval, Ratio) with suitable real-world examples. [14]
6. The following data represents chemical impurity (ppm) in 25 water samples:

11, 19, 24, 30, 12, 20, 25, 29, 15, 21, 24, 31, 16, 23, 25, 26, 32, 17, 22, 26, 35, 18, 24, 18, 27

Construct a continuous frequency distribution table with class interval width of 5 and draw a Histogram. [14]
7. What are the measures of Central Tendency? Explain methods for calculating Arithmetic Mean for grouped and ungrouped data. [14]
8. Based on the 5-point summary below, construct a Box and Whisker Plot. Is the distribution symmetrical/normal? Give reasons. [14]

Minimum	Q1 (First Quartile)	Q2 (Median)	Q3 (Third Quartile)	Maximum
5	12	19	26	50

[14]

9. Write short notes on any TWO of the following: [14]

- (i) Graphical representation of data in Excel
- (ii) Uses of Frequency Distribution in data analysis
- (iii) Pivot Charts and Dashboard creation
- (iv) Analysis of Variance (ANOVA) in Excel